

# The audit

## DAVIS (primary)

30,056 pairs · 68 drugs · 442 targets  
4 levels: random, cold-drug,  
cold-target, cold-pair

## KIBA (replication)

118,254 pairs · 2,111 drugs · 229 targets  
2 levels: random, cold-drug  
(422 held-out drugs vs DAVIS's 13)

## Retrained on identical splits — 3 seeds per cell

ColdSite-DTI (ours) · HyperAttentionDTI · MolTrans + DeepDTA (accuracy anchor, no attention)  
48 DAVIS cells · 18 KIBA cells · every cell scored by AUROC before its explanation is read

## The explanation, read three ways

attention (as published) · integrated gradients on the same weights · uniform map = the floor  
alternative attention readouts test whether a verdict belongs to the model or to the reduction

## PLAUSIBILITY — does it point there?

precision@10 against  
· UniProt annotated residues  
· KLIFS 85-residue ATP pocket  
· the drug's own crystal contacts  
nulls: borrowed map, same amino acid,  
within the site-spanning stretch,  
60 unseen non-kinase proteins

## FAITHFULNESS — is it used?

mask the top-10 and re-predict,  
against a control of the same size  
in the space each model reads  
(residues; tokens for MolTrans)  
  
 $\text{delta} > 0$  = load-bearing

## One correction per arm, and a control for the metric itself

Holm-Bonferroni once over the whole family: 16 DAVIS cells, 6 KIBA cells, 12 for the gradient  
positive control: a planted explanation of known dose is detectable at 2% — so a null is a null, not a weak test  
bootstrap CIs over proteins · every cell reported with its three seeds